

Characterization of projective toric varieties from dynamical viewpoints

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Contents

1. Background & Motivation
2. Previous results
3. Main results & Proofs
4. Further discussion & Open questions

Based on the joint works with Jia Jia, Sheng Meng and De-Qi Zhang.

Background & Motivation

- We start from a smooth projective variety X over $\mathbb{C}$.
- Suppose X admits an endomorphism $f : X \rightarrow X$ of degree > 1 .

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- ($\dim X = 1$): Hurwitz formula $\Rightarrow X$ is either rational or elliptic.
- ($\dim X = 2$): Such X (including the singular case) has been fully studied by Nakayama.
- ($\dim X \geq 3$): There are three (equivariant) typical fibrations to study such X : **Albanese map**, **Kodaira fibration**, and **maximal rationally connected fibration** (MRC).

$$f \circ X \dashrightarrow Y \circ g$$

Background & Motivation

Equivariant Minimal Model Program (EMMP)

Let X be a normal projective variety (mildly singular) admitting an endomorphism f . Given any finite sequence of MMP, we say that it is **f -equivariant** if the following diagram commutes for some iteration $f_1 = f^s$:

$$\begin{array}{ccccccc} X = X_1 & \dashrightarrow & X_2 & \dashrightarrow & \cdots & \dashrightarrow & X_r \\ \downarrow f_1 & & \downarrow f_2 & & & & \downarrow f_r \\ X = X_1 & \dashrightarrow & X_2 & \dashrightarrow & \cdots & \dashrightarrow & X_r \end{array}$$

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Theorem (Existence of the EMMP)

EMMP holds in the following situations:

- X is a Mori dream space (e.g. Fano varieties, toric varieties).
- (Nakayama) $\dim(X) = 2$ and $\deg(f) > 1$.
- (Meng-Zhang) X admits a polarized or int-amplified endomorphism.

Background & Motivation

Definition (Meng-Zhang, 2018; Meng, 2020)

Let $f : X \rightarrow X$ be a surjective endomorphism of a projective variety X .

- (i) f is **polarized**, if $f^*H \sim qH$ for some ample H and integer $q > 1 \Leftrightarrow f^*|_{N^1(X)}$ is diagonalizable with all eigenvalues being of modulus $q > 1$, where $N^1(X) = NS(X) \otimes_{\mathbb{Z}} \mathbb{R}$.
- (ii) f is **int-amplified**, if $f^*H - H$ is ample for some ample $H \Leftrightarrow$ all the eigenvalues of $f^*|_{N^1(X)}$ are of modulus > 1 . **Polarized $\Rightarrow$ Int-amplified**

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Why polarized and int-amplified?

- Eigenvalue properties of $f^*|_{N^1(X)}$, to eliminate the non-essential case: the product of an automorphism and an endomorphism.
- “Int-amplified” is preserved under product map: e.g. $\mathbb{P}^1 \times \mathbb{P}^1$.

Background & Motivation

Definition (Toric varieties)

A normal projective variety X of dimension n is a **toric variety** if X contains a big torus $T = (k^*)^n$ as an (affine) open dense subset such that the natural multiplication action of T on itself extends to an action on the whole variety (equivariant embedding). We say $(X, X \setminus T)$ is a **toric pair**.
When X is $\mathbb{Q}$ -factorial, $\#$ components of $X \setminus T = \dim(X) + \rho(X)$.

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Question A: (Fakruddin, 2003 (resp. Meng-Z., 2020))

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Let X be a smooth projective rationally connected variety admitting a polarized (resp. int-amplified) endomorphism. Is X a toric variety?

Question B: (A special case of Question A)

Let X be a smooth Fano variety of Picard number one. Suppose X admits a non-isomorphic (and hence polarized) endomorphism f . Is $X \cong \mathbb{P}^n$?

Previous results (related to the main questions)

Theorem (Nakayama, 2002 (Sato's conjecture))

A smooth rational surface is toric if and only if it admits a non-isomorphic surjective endomorphism. (Rmk: ✓ Question A in dimension ≤ 2)

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Theorem (Amerik-Rovinsky-Van de Ven, 1999; Hwang-Mok, 2003)

Let X be a smooth Fano threefold of Picard number one which admits a non-isomorphic endomorphism. Then $X \cong \mathbb{P}^3$. (Rmk: ✓ Question B in dimension ≤ 3)

Previous results (toric structures associated to fibrations)

Proposition (splitting $\mathbb{P}^1$ -bundle over toric varieties)

Let $X := \mathbb{P}_Y(\mathcal{E}) \xrightarrow{\tau} Y$ be a splitting $\mathbb{P}^1$ -bundle over a normal projective variety Y . Suppose (Y, Δ_Y) is a toric pair. Then X is toric, and for any two disjoint sections S_0, S_1 of τ , $(X, S_0 + S_1 + \tau^{-1}(\Delta_Y))$ is a toric pair.

- A fibration $\tau : X \rightarrow Y$ of smooth projective varieties is a *conic bundle* if every fibre is isomorphic to a conic.
- A conic bundle $\tau : X \rightarrow Y$ is a (*smooth*) $\mathbb{P}^1$ -*bundle*, if τ is smooth (and every fibre is isomorphic to $\mathbb{P}^1$).
- A $\mathbb{P}^1$ -bundle $\tau : X \rightarrow Y$ is an *algebraic $\mathbb{P}^1$ -bundle*, if $X \cong \mathbb{P}_Y(\mathcal{E})$ for some locally free rank-two sheaf $\mathcal{E}$ on Y .
- An algebraic $\mathbb{P}^1$ -bundle $X := \mathbb{P}_Y(\mathcal{E}) \xrightarrow{\tau} Y$ is a *splitting $\mathbb{P}^1$ -bundle* if $\mathcal{E}$ is a direct sum of two invertible sheaves.

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How to relate this to self-morphisms?

Previous results (toric structures associated to fibrations)

Theorem (Amerik, 2003)

Let $X = \mathbb{P}_Y(\mathcal{E}) \xrightarrow{\tau} Y$ be an algebraic $\mathbb{P}^1$ -bundle over a smooth projective rational variety Y . If X admits an endomorphism which is not of degree one on a general fibre, and $\mathcal{E}$ is H -semistable for some ample divisor H , then τ is a splitting $\mathbb{P}^1$ -bundle.

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When Y is the projective space, Amerik further proved the following (actually holds true for the base Y being simply connected and the first cohomology groups of all line bundles on Y are zero).

Theorem (Amerik, 2003)

Let $Y := \mathbb{P}^n$ and $X := \mathbb{P}_Y(\mathcal{E})$ with $\mathcal{E}$ a vector bundle of rank two. Then X admits a non-isomorphic endomorphism if and only if $\mathcal{E}$ is a direct sum of two line bundles.

The above two theorems have been extended to higher rank vector bundles by Amerik and Kuznetsova in 2017.

Previous results (totally invariant ramifications)

Theorem (Hwang-Nakayama, 2007)

Let X be a Fano manifold of Picard number one different from the projective space. If an endomorphism of X is étale outside a totally invariant divisor, it is bijective.

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Theorem (Hwang-Nakayama, 2007)

Let X be a Fano manifold of Picard number one different from the projective space. If an endomorphism of X is étale outside a totally invariant divisor, it is bijective.

- Therefore, if there is a non-isomorphic $f : X \rightarrow X$ and a reduced divisor D such that $f^{-1}(D) = D$ and $f|_{X \setminus D} : X \setminus D \rightarrow X \setminus D$ is étale, then X is a projective space. (✓ Question B if f has totally invariant ramifications)

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- Therefore, if there is a non-isomorphic $f : X \rightarrow X$ and a reduced divisor D such that $f^{-1}(D) = D$ and $f|_{X \setminus D} : X \setminus D \rightarrow X \setminus D$ is étale, then X is a projective space. (✓ Question B if f has totally invariant ramifications)
- We extended this result to the case $\rho(X) \geq 2$ to study Question A.

Theorem (Meng-Zhang, 2019 (resp. Meng-Z., 2020))

Let X be a rationally connected smooth projective variety with D a reduced divisor. Then (X, D) is a toric pair if and only if X admits a polarized (resp. int-amplified) endomorphism f such that $f|_{X \setminus D}$ is étale.

Main results — Threefolds

Theorem A (Meng-Zhang-Z., 2020)

A smooth Fano threefold is either toric or a product of $\mathbb{P}^1$ and a del Pezzo surface if and only if it admits a non-isomorphic surjective endomorphism.

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- Del Pezzo surfaces being rational, Theorem A gives a corollary that a smooth Fano threefold admitting a non-isomorphic endomorphism is rational, which was first proved by Zhang in 2012.
- Theorem B gives a positive answer to Question A when X is a smooth Fano threefold.
- Our proof is classification-free, and analysing conic bundle structures is the main tool in this project.

Facts on the conic bundle structures

- A fibration $\tau : X \rightarrow Y$ of smooth projective varieties is a **conic bundle** if every fibre is isomorphic to a conic, i.e., a scheme of zeros of a nonzero homogeneous form of degree 2 on $\mathbb{P}^2$.
- $\Delta_\tau := \{y \in Y \mid \tau \text{ is not smooth over } y\}$ is the **discriminant** of τ , which is a reduced divisor (Sarkisov, 1983).
- A conic bundle $\tau : X \rightarrow Y$ is said to be **elementary**, if $\rho(X/Y) = 1$.
- Any three irreducible components of Δ_τ have no common intersection (Sarkisov, 1983 and Jia-Z., 2020).

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- Any three irreducible components of Δ_τ have no common intersection (Sarkisov, 1983 and Jia-Z., 2020).
- A conic bundle τ has **(not unique)** factorization: $X \rightarrow \cdots \rightarrow X_r \rightarrow Y$ with $\tau_r : X_r \rightarrow Y$ being elementary, $X_i \rightarrow X_{i+1}$ being blow-ups of smooth manifolds M of codimension two, and $\Delta_\tau = \Delta_{\tau_r} \amalg M$.
- When Y is a surface and $\Delta_\tau \neq \emptyset$, if y is a smooth point of Δ_τ , then X_y is a reducible conic in $\mathbb{P}^2$; if y is a double point of Δ_τ , then X_y is a double line in $\mathbb{P}^2$.

Proofs for the main results (Threefold) - - Conic bundle

Let $f : X \rightarrow X$ be a non-isomorphic surjective endomorphism on a smooth Fano threefold.

- If $\rho(X) = 1$, then $X \cong \mathbb{P}^3$ (Amerik-Rovinsky-Van de Ven 1999, Hwang-Mok 2003).

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- If $\rho(X) = 2$, then we show that either $X \rightarrow \mathbb{P}^2$ admits a Fano contraction, or $X \rightarrow \mathbb{P}^3$ is a toric blow-up.

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- If $\rho(X) \geq 3$, then X admits a (not necessarily elementary) conic bundle structure (Mori-Mukai 1983 + dynamical assumption).

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- If $\rho(X) \geq 3$, then X admits a (not necessarily elementary) conic bundle structure (Mori-Mukai 1983 + dynamical assumption).
- If $\tau : X \rightarrow Y$ is a fibre-dimension one Fano contr., then we show that

$\tau \xRightarrow{(i)}$ conic bundle $\xRightarrow{(ii)}$ (smooth) $\mathbb{P}^1$ -bundle, i.e., $\Delta_\tau = \emptyset$

$\xRightarrow{(iii)}$ algebraic $\mathbb{P}^1$ -bundle $\mathbb{P}_Y(\mathcal{E})$

$\xRightarrow{(iv)}$ splitting $\mathbb{P}^1$ -bundle $\mathbb{P}_Y(\mathcal{O} \oplus \mathcal{L})$ (if f is further IAMP)

Proofs for the main results (Threefold) - - Conic bundle

(i) Fano contr. of rel dim one $\implies$ (elementary) conic bundle

Mori's (1982) motivating result (for threefold). In general, Ando (1985) showed that if the Fano contr. (from a smooth projective variety) is equi-dimensional of rel dim one, then it is a conic bundle.

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(ii) elementary conic bundle $\implies$ smooth $\mathbb{P}^1$ -bundle

This step is too technical and we skip its proof here. The main idea is to assume by contradiction that the discriminant $\Delta_\tau \neq \emptyset$. Then by running the equivariant minimal model program, we reduce to the case when Y is $\mathbb{P}^2$ or $\mathbb{P}^1 \times \mathbb{P}^1$. This step is closely related to the dynamics.

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(iii) smooth $\mathbb{P}^1$ -bundle $\implies$ algebraic $\mathbb{P}^1$ -bundle (see Elencwajg 1981)

Let $\tau : X \rightarrow Y$ be a (smooth) $\mathbb{P}^1$ -bundle over a smooth projective rational variety Y . Then $X = \mathbb{P}_Y(\mathcal{E})$ for some locally free rank-two sheaf $\mathcal{E}$ over Y .

Proofs for the main results (Threefold) - - Conic bundle

splittingness of algebraic $\mathbb{P}^1$ -bundle (I: Elementary case)

(Amerik, 2003) Let $\tau : X = \mathbb{P}_Y(\mathcal{E}) \rightarrow Y$ be an algebraic $\mathbb{P}^1$ -bundle over a smooth projective rational variety. Suppose $\text{rank } \mathcal{E} = 2$ and X admits a non-isomorphic endomorphism f . Then τ splits if either

- $\deg f > \deg f|_Y$ (this is the case when f is IAMP) and $\mathcal{E}$ is H -semistable for some ample divisor H ; or
- $Y = \mathbb{P}^n$.

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splittingness of algebraic $\mathbb{P}^1$ -bundle (II: Birational dominance case)

Let X be a smooth Fano threefold admitting an IAMP endomorphism f . Suppose that X admits a conic bundle $\tau : X \rightarrow Y$, which factors as $X \xrightarrow{\pi} W \xrightarrow{\tau_0} Y$ with τ_0 being a (smooth) $\mathbb{P}^1$ -bundle. Then τ_0 is a splitting $\mathbb{P}^1$ -bundle. (Here, W is not necessarily Fano!)

Proofs for the main results (Threefold) - - Conic bundle

Equivariant minimal model program for conic bundle structures

Let X be a smooth Fano threefold with a conic bundle $\tau : X \rightarrow Y$, and $f : X \rightarrow X$ a non-isomorphic endomorphism. Then there exists an f -equivariant MMP

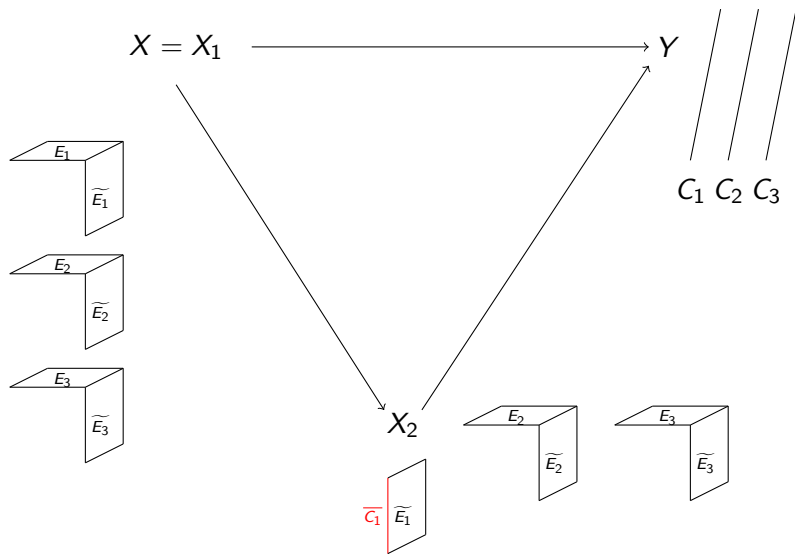
$$X = X_1 \rightarrow \cdots \rightarrow X_i \rightarrow \cdots \rightarrow X_{r+1} \rightarrow Y$$

such that the following hold.

- ① $r = \rho(X) - \rho(Y) - 1$ and each X_i is a smooth projective threefold.
- ② $\tau_{r+1} : X_{r+1} \rightarrow Y$ is an algebraic $\mathbb{P}^1$ -bundle $\mathbb{P}_Y(\mathcal{E})$ over Y .
- ③ $\pi : X \rightarrow X_{r+1}$ is the blowup of X_{r+1} along r disjoint union of $(f|_{X_{r+1}})^{-1}$ -invariant smooth curves $\bigcup_{i=1}^r \overline{C}_i$ with $\tau_{r+1}(\overline{C}_i) = C_i$.

If f is further assumed to be int-amplified, then we can show there is a suitable route such that X is a toric blow-up of a not necessarily Fano splitting $\mathbb{P}^1$ -bundle over a toric surface, in which case, X is toric.

Proofs for the main results (Threefold) - - Conic bundle



Proofs for the main results (Threefolds)

Theorem (Reduce to the int-amplified case (skip the proof))

Let f be a non-isomorphic endomorphism of a smooth Fano threefold X . Then either f is int-amplified after iteration or $X \cong Y \times \mathbb{P}^1$ for a del Pezzo surface Y .

Note that toric del Pezzo surfaces are: $\mathbb{P}^2$, $\mathbb{P}^1 \times \mathbb{P}^1$, $\text{Bl}_1\mathbb{P}^2$, $\text{Bl}_2\mathbb{P}^2$ or $\text{Bl}_3\mathbb{P}^2$.

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Theorem (together with the case $\rho(X) \leq 2 \Rightarrow$ our main result)

Let X be a smooth Fano threefold with $\rho(X) \geq 3$. Suppose that X admits an IAMP endomorphism f . Then X is toric.

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Idea for the proof of the above theorem

The point is to show X is a (toric) blow-up of a (not necessarily Fano) splitting $\mathbb{P}^1$ -bundle over a toric surface along disjoint curves which are intersections of several f^{-1} -invariant prime divisors. By **the splittingness of algebraic $\mathbb{P}^1$ -bundle**, our theorem follows.

Proofs for the main results (Threefold)

Why we need to blow up curves which are intersections of totally invariant prime divisors?

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Why we need to blow up curves which are intersections of totally invariant prime divisors?

Reason (Theorem, Meng-Zhang-Z., 2020):

Let $\tau : X \rightarrow Y$ be a splitting $\mathbb{P}^1$ -bundle over a smooth projective rational surface Y . Suppose that X admits an IAMP endomorphism f . Then X is toric and there is a toric pair (X, Δ) such that Δ contains the union Σ of all the f^{-1} -periodic prime divisors.

Then we can make the birational part of the factorization of a Fano conic bundle be the blow-ups of toric boundaries, in which case, X is toric.

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Why we need to blow up curves which are intersections of totally invariant prime divisors?

Reason (Theorem, Meng-Zhang-Z., 2020):

Let $\tau : X \rightarrow Y$ be a splitting $\mathbb{P}^1$ -bundle over a smooth projective rational surface Y . Suppose that X admits an IAMP endomorphism f . Then X is toric and there is a toric pair (X, Δ) such that Δ contains the union Σ of all the f^{-1} -periodic prime divisors.

Then we can make the birational part of the factorization of a Fano conic bundle be the blow-ups of toric boundaries, in which case, X is toric.

A further question

What about the totally periodic subvarieties of higher codimension? Can we find a toric pair such that the boundary divisor contains all of such subvarieties?

Further discussion — Fourfolds

Theorem (Jia-Z., 2020)

Let X be a smooth Fano fourfold admitting a conic bundle. Then X is toric if and only if X admits a polarized (or IAMP) endomorphism.

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- The appearance of small contr. makes things much more complicated.
- We are not able to deal with the case of Fano fourfolds of Picard number one admitting non-isomorphic endomorphisms.
- The Fano contr. from a Fano fourfold may not be equi-dimensional, even if it is of relative dimension one ($\nrightarrow$ a conic bundle).
- Can we eliminate the special fibre using the dynamical condition?
- We start from a (not necessarily elementary) conic bundle τ .
- We show that (i) there is an EMMP for τ , (ii) a suitable elementary conic bundle under the dynamic assumption is a splitting $\mathbb{P}^1$ -bundle, and (iii) X is a toric blow-up of some splitting $\mathbb{P}^1$ -bundle over a toric Fano threefold.

Further discussion — Higher dimensional manifolds

- If $X \rightarrow Y$ is a conic bundle with either all the fibres being reduced or $\dim Y \leq 3$, and X is Fano, then Y is also Fano (Wiśniewski, 1991).
- There is no classification for Fano manifolds of dimension ≥ 4 ; it has been intensively studied by Casagrande, Romano, Montero, Noce etc. in terms of the Lefschetz defect:

$$\delta_X := \max\{\dim \ker(H^2(X, \mathbb{R}) \rightarrow H^2(D, \mathbb{R})) \mid D \text{ is a prime divisor} \}$$

which lies in $\{0, 1, 2, \dots, \rho_X - 1\}$.

- The case $\delta_X \geq 3$ is now well-understood. Casagrande showed that if $\delta_X \geq 4$, then X splits into a product of Fano surface and a Fano manifold. Montero and Romano showed that if $\dim X = 4$ and $\delta_X = 3$, then X admits a conic bundle. They generalized it to higher dimensional case (to appear). The case $\delta_X \leq 2$ is less known.
- Toric Fano manifolds are classified up to dimension four. There exists a toric Fano fourfold, with all the contractions being small.

Open questions

Conjecture (totally invariant subvarieties)

Let $f : X \rightarrow X$ be a non-isomorphic endomorphism on the projective space $X = \mathbb{P}^n$. Let Z be an f^{-1} -periodic closed subvariety. Is Z linear?

The above conjecture has been confirmed in the following cases:

- $X = \mathbb{P}^2$ (Gurjar 2003).
- $X = \mathbb{P}^3$ and $\dim Z = 2$ (Höring 2017, Nakayama-Zhang 2010).
- Z is a smooth hypersurface of X (Cerveau-Lins Neto 2000, Beauville 2001, Gurjar 2003).
- Z is a hypersurface with isolated singularities (MABED Yanis, 2021)

A positive answer to the above conjecture in $\dim X = 4$ will help us solve our toric problem completely in $\dim X = 4$.

Open questions

Question (toric boundary conjecture)

Let $f : X \rightarrow X$ be an IAMP (or polarized) endomorphism of a rationally connected smooth projective variety X . Let Σ be the union of f^{-1} -periodic divisors. Is X toric, and if so, is there a toric pair (X, Δ) such that $\Sigma \subseteq \Delta$? (Meng-Zhang, 2019) There are only finitely many totally periodic subvarieties when f is IAMP.

The above question has a positive answer in the following cases:

- $\dim X = 2$ (Meng-Zhang-Z., 2020).
- X is a smooth Fano threefold (Meng-Zhang-Z., 2020, Jia-Z., 2020).
- Higher codimensional case will be much harder. For example, does $\mathbb{P}^2$ have at most three totally invariant points?
- Can we give an (optimal) upper bound for $\#\Sigma$? This question indeed makes sense for arbitrary normal projective varieties.

Thank You!